

Rebuilding the POTS Ontology from Seven Seed Papers

Ontology first, evidence first: method, outputs and results

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Summary

We rebuilt the literature layer for a computable phenotype of postural orthostatic tachycardia syndrome (POTS) from the beginning. We used seven seed papers, two of them 2026 reviews (Lau et al., *Heart Lung Circ*; Chung & Raj, *JAMA*), and an *evidence-first* method. First we wrote down the questions the ontology must answer. Then we read every seed paper and recorded each relevant claim as a verbatim quote. Only after that did we create topics (*facets*) and links, both generated from those claims. The vocabulary, the PubMed harvest, the tagging, the concept mapping, the report and the viewer were then all regenerated from the new ontology.

Seed papers	7	Facets (topics)	72	Articles harvested	2,668
Claims extracted	407	Mechanism flags	7	Facet tag rows	10,671
Quoted passages kept	401	Literature links	132	Candidate concepts	308
Quotes verified verbatim	412/412	Taxonomy edges	102	Report checks	23/23
Passages against a link	5	Classes	38	Tests	116 pass

Table 1: Headline numbers. Run `run_20261001T190057Z`, profile `core`.

1 Method

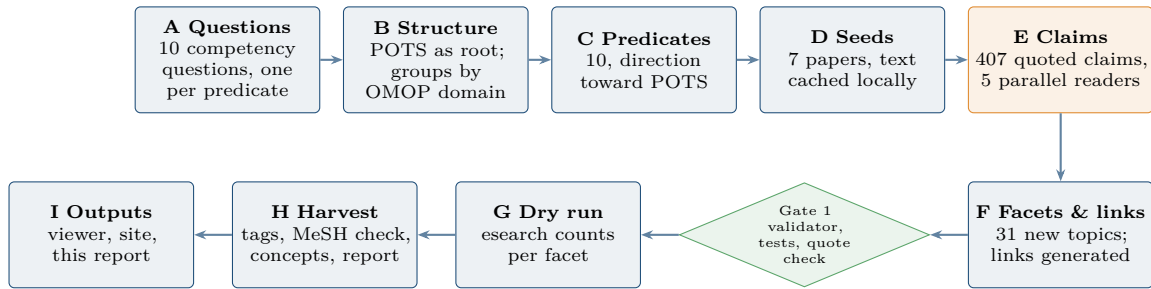


Figure 1: The sequence followed. The orange step is where the evidence enters: no link was written that did not start as a quoted claim.

Step	What was done	Output
A Questions	Ten competency questions: causes of POTS, tests showing a cause, treatments implying a cause, treatments for POTS, co-occurring conditions, triggers, conditions to rule out, what a test measures, battery components, kind of topic	<code>docs/COMPETENCY_QUESTIONS.md</code>
B Structure	POTS is the root of the evidence graph; links point from tests and treatments to causes, and from causes and conditions to POTS. Topics are grouped by the OMOP domain they will be observed in	4 facet groups; 38 classes
C Predicates	Ten predicates, each tied to one question; only <code>is_a</code> is subsumption	<code>relations.yaml</code> predicates
D Seeds	Seven papers. Full text for six (PMC open access for four; the 2026 PDFs for Lau and Chung); abstract only for Larsen. Texts kept locally, not committed	<code>seed_references</code>
E Claims	Each paper was read in full by a separate reader working to the same instructions: the claim, a verbatim quote of one to two sentences, the section, and a verdict (supports or partial)	407 claims
F Facets & links	Proposed topics were merged or dropped (§2.1). The links in <code>relations.yaml</code> are generated from the claims by <code>scripts/build_relations.py</code> ; taxonomy edges are kept by hand	72 facets, 132 links
Gate 1	Vocabulary validator; test suite, including a test that every literature link has a backing claim and that the generated block is current; every quote checked against the source text	all pass
G Dry run	research count for every facet within the corpus query, before harvesting	no facet with zero hits
H Harvest	Full corpus harvest and facet tagging, MeSH descriptor check, concept resolution, validation report	2,668 articles; 23/23 checks

Table 2: Each step and its output.

2 Seed papers and claims

Key	Paper	Source text	Access	Claims	Supports	Partial
lau2026	Lau et al., State-of-the-art review, <i>Heart Lung Circ</i> 2026 (PMID 41519610)	PDF full text	CC BY	98	49	49
chung2026	Chung & Raj, Review, <i>JAMA</i> 2026 (PMID 42635998)	PDF full text	AMA, rights reserved	104	62	42
low2009	Low et al., <i>J Cardiovasc Electrophysiol</i> 2009	PMC3904426	open access	97	55	42
angeli2024	Angeli et al., <i>Sci Rep</i> 2024	PMC10761725	open access	43	26	17
gibbons2013	Gibbons et al., <i>PLoS ONE</i> 2013	PMC3874039	open access	26	15	11
zhang2022	Zhang et al., <i>Cureus</i> 2022	PMC9629858	open access	23	10	13
larsen2025	Larsen et al., long-COVID POTS, medRxiv	abstract	preprint	16	6	10
				407	223	184

Table 3: Seed papers. The JAMA paper’s quotes are kept short; the repository is private and the public site withholds them.

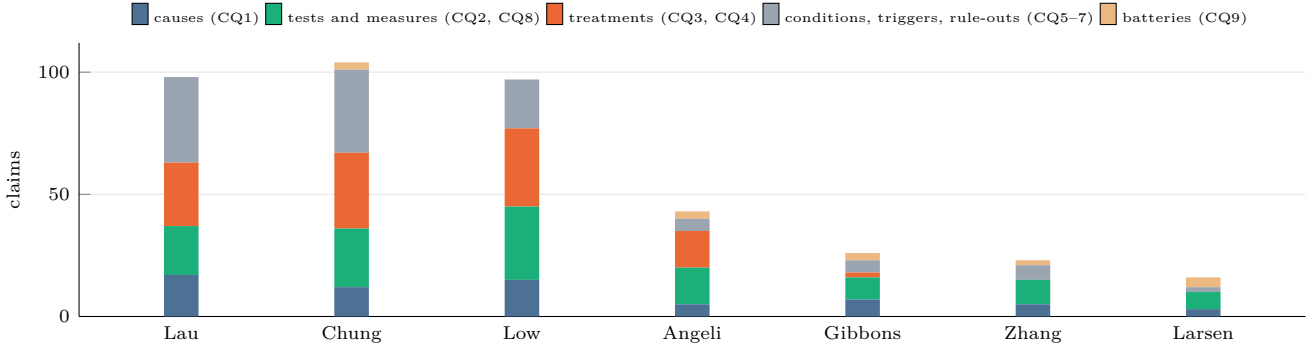


Figure 2: Claims per paper by the question they answer. The two 2026 reviews contribute half the claims and most of the triggers, rule-out conditions and treatments; the cohort studies contribute mostly test evidence.

Quote verification. `scripts/verify_excerpts.py` checks every quote against the cached source text after collapsing whitespace, folding typographic quotes and dashes, and rejoining words hyphenated across PDF line breaks. All 412 quotes are found (401 kept on links, 11 in claims not turned into links). Twelve match only once PDF word spacing and hyphenation are ignored: the PDF text layer had dropped spaces between words. The displayed quotes restore those spaces from a second text extraction (`pdfotext`).

2.1 Turning proposed topics into facets

Readers proposed 51 new topic ids where a claim had no matching facet. 25 became facets as proposed, 17 were merged into 11 facets (6 of them newly named, e.g. “other causes of sinus tachycardia”), and 9 were dropped: 31 new facets in all. The decisions are recorded in `extraction/2026-10-01/topic_map.yaml`. The dropped claims are kept under `not_added` with the reason.

Decision	Topics
Added: mechanism (1)	cerebral hypoperfusion (named as a proposed mechanism by Chung 2026 and Lau 2026)
Added: tests (7)	24-h urine sodium; plasma renin and aldosterone; regional noradrenaline spillover; muscle sympathetic nerve activity (microneurography); cardiac MRI; quantitative sensory testing; end-tidal CO ₂ on tilt
Added: conditions to rule out (4)	other causes of sinus tachycardia (merged from anaemia, endocrine, transient-state and secondary-tachycardia proposals); arrhythmias and inappropriate sinus tachycardia; structural heart and chronic lung disease (merged); medication-induced tachycardia (merged)
Added: onset triggers (5)	trauma or concussion (merged); surgery (merged); hormonal change (merged from pregnancy and hormonal proposals); adolescent growth spurt; prolonged bed rest
Added: co-occurring (7)	anxiety and depression (merged with somatic hypervigilance); fibromyalgia; sleep disturbance; iron deficiency; autism and ADHD; endometriosis; functional neurological disorder
Added: treatments (7)	physical countermeasures; behavioural and dietary measures; stopping drugs that worsen POTS; auricular vagus nerve stimulation; antihistamines; prokinetics; phenobarbital
Merged into existing	irritable bowel syndrome → GI dysmotility (relabelled to include functional GI disorders)
Dropped (9)	central neuroinflammation; microvascular dysfunction (hypothetical, one paper); COMPASS-31 and the Malmö POTS score (symptom scores, not mechanism evidence); five-point hypermobility score; exclusion laboratory work-up; lower-body negative pressure (research test); acquired neuropathy causes; diabetes

Table 4: Topic decisions. Existing facet definitions were also corrected where the seeds disagreed with them: the neuropathic deficit is often patchy rather than length-dependent, and reduced nerve fibre density is reported in 24% to 45–53% of patients biopsied.

3 The ontology

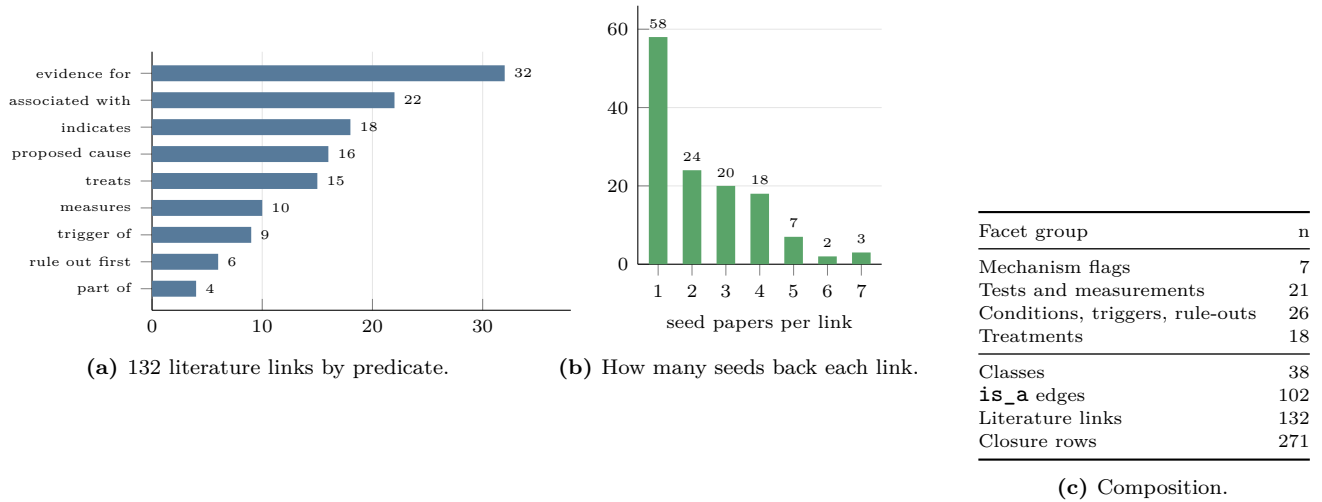


Figure 3: Shape of the rebuilt ontology. 58 links rest on a single seed paper; 50 are backed by three or more. 49 links have only partly supporting passages.

Mechanism flag	Tests	Treatments	Conditions	Causes	Articles (query)	(MeSH)	(any)
Neuropathic	9	5	4	2	235	169	344
Hyperadrenergic	5	4	1	1	226	139	277
Hypovolemic	5	4	1	1	177	49	188
Deconditioning	4	1	3	0	105	42	129
Autoimmune	1	1	1	2	186	53	190
Post-infectious	0	0	0	0	334	157	338
Cerebral hypoperfusion	1	0	0	0	181	73	209

Table 5: What points to each cause, and how many harvested articles carry it. Neuropathic is by far the best characterised; post-infectious is linked to POTS and to other causes but no test or treatment points to it.

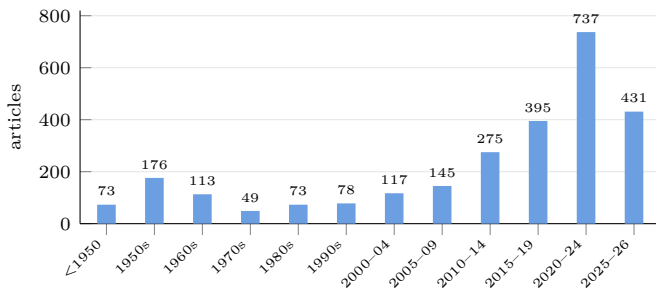
Contested link	Against	Passage
hypovolemic → proposed cause of POTS	Low 2009	“The term idiopathic hypovolemia is unsatisfactory, since most patients do not have reduced plasma volumes or red cell mass.”
deconditioning → proposed cause of POTS	Lau 2026	“...suggest that deconditioning is not a primary causative factor”
cerebral hypoperfusion → proposed cause of POTS; brain blood flow → evidence for it	Larsen 2025	“Measures of cerebrovascular changes during orthostatic challenge were also similar and failed to correlate with symptoms of orthostatic intolerance.”
SSRI / central agents → treats POTS	Lau 2026	“Others have found the use of serotonin-norepinephrine reuptake inhibitor to worsen POTS symptoms and orthostatic tachycardia”

Table 6: Links kept because at least one seed supports them, but with a passage arguing against. They are flagged in the viewer.

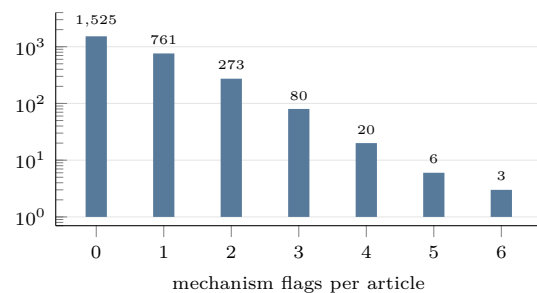
4 Gates and harvest

Check	Detail	Result
Vocabulary validator	72 facets, 110 nodes, 234 edges, 10 predicates	clean
Test suite	incl. every link has a backing claim; generated block current; seven flags each linked to POTS; no flag is_a POTS	116 pass
Quote check	against the cached seed texts	412/412
Dry run	esearch count per facet before harvesting; lowest: stopping offending drugs (1), growth spurt (2), prokinetics (3)	0 empty facets
MeSH descriptors	224 declared; all recognised by PubMed; 29 unused in this corpus	0 invalid
Harvest	esearch 2,668, retrieved 2,668, 101 E-utilities requests	complete
Seed resolution	six seeds found by title; Larsen absent from PubMed as expected	7/7
Report checks	corpus, seeds, facet coverage, graph integrity, MeSH validity	23/23

Table 7: Gates passed before and after harvesting.



(a) Corpus by publication year (6 undated).



(b) Mechanism co-occurrence (seven flags).

Figure 4: Corpus. 1,143 articles (42.8%) carry at least one mechanism flag and 382 carry two or more. Of facet tags, only 25.0% are found by both the title/abstract query and NLM’s MeSH indexing (2,136 both; 4,436 query only; 1,963 MeSH only).

Composition	n	%
Articles	2,668	100.0
With abstract	1,950	73.1
MeSH indexed	2,015	75.5
Human check-tag	1,940	72.7
Has PMC id	1,048	39.3
Non-English	425	15.9
Historical clause only	555	20.8

Concepts	Candidates	Facets	Loose best
SNOMED (OLS4)	208	54	
LOINC (NLM CTS)	68	17	6
RxNorm (RxNav)	32	12	
Exact 89 / contains 190 / loose 29; all unreviewed			
9 declared fallback gaps (absent from OLS4’s SNOMED subset)			

Table 8: Corpus composition (left) and candidate OMOP concepts (right). No `concept_id` values yet: ATHENA needs an interactive login.

5 Outputs

Output	Content
docs/COMPETENCY_QUESTIONS.md	The ten questions and the predicate that answers each
config/relation_excerpts.yaml	Source of truth for literature links: 132 links, 401 passages, 11 claims not added
config/relations.yaml	102 hand-kept <code>is_a</code> edges and the generated block of 132 links
config/facets.yaml	72 facets with queries, MeSH descriptors, concept search terms and seed references
scripts/build_relations.py, verify_excerpts.py	Generate links from claims; check quotes against local seed texts
data/articles.jsonl, data/derived/*.tsv	The 2,668-article corpus and the derived tables
docs/HARVEST_REPORT.md	Validation report for the run
reports/ontology_graph.html	Evidence map: a DAG rooted at POTS, a map, an outline, every link with its passages and verdicts
Public site	https://jeonglab-bcm.github.io/pots-phenotyping-site/ , published by <code>scripts/publish_site.sh</code> from this private repository; JAMA quotes withheld

Table 9: What the rebuild produced.

6 Limitations and next steps

- **Unreviewed judgements.** Quotes are verbatim, but which passage backs a claim, and whether it fully or only partly supports it, were judged by an AI reader. 49 links rest only on partial support. A clinician review with the viewer’s CSV export is the next gate.
- **Review-heavy evidence.** The two 2026 reviews supply half the claims; 58 links rest on a single paper. The weakest branches are post-infectious and cerebral hypoperfusion, which have no test or treatment links, and cerebral hypoperfusion is also contested.
- **Abstract only for Larsen 2025**, so its claims are short and mostly partial.
- **Concepts are candidates**, with no `concept_id` until an ATHENA bundle is loaded; 6 best candidates are loose matches.
- **Keyword tagging** finds papers that discuss a topic, not papers that measured it; query and MeSH agree on only a quarter of tags.
- **Licence.** The JAMA paper reserves text and data mining rights. Its text is not committed, quotes are short, and the public site withholds them.